International Journal of Computing and Digital Systems

When Do Text Proxies Suffice? A Regime Analysis of Sparse, Dense, and Native-Visual Retrieval Under Proxy Noise

Abstract: Multimodal retrieval systems often index textual proxies, such as OCR output or generated captions, instead of raw visual media. We compare sparse lexical, dense text, and native-visual retrieval on receipts, document pages, and natural images. For verbatim receipt lookup, character 3–5gram BM25 outperforms word BM25 on Apple Vision clean OCR (MRR 0.9436 vs. 0.8371), degraded OCR (0.8248 vs. 0.6972), and Tesseract clean OCR (0.8412 vs. 0.7364). Under synthetic entity-substituting paraphrases, this ordering reverses (0.2427 vs. 0.4334 on `gpt-4o-mini`), whereas under open-weight `Qwen2.5-7B` and human queries where topical anchors are preserved, character and word BM25 perform comparably (0.5802 vs. 0.6292 and 0.5549 vs. 0.5766). ColQwen2 achieves MRR 0.4962 on the primary synthetic paraphrase set (compared with 0.4334 for word BM25), but was not evaluated across secondary or human query sets. On a 500-query DocVQA subset, ColQwen2 reaches MRR 0.6450, while tested text-proxy baselines reach at most 0.3507. On 1,000 MS-COCO images, SigLIP reaches MRR 0.6071, outperforming baseline BLIP captions (0.3728) and detailed Florence-2 captions (0.4420; all $p < 10^{-4}$). Query formulation strongly affects dense retrieval: BGE-Base MRR rises from 0.3256 on short snippets to 0.5263 on natural answerable questions. Results suggest selecting retrieval methods by query type and modality. Generalization is limited by sampled benchmark subsets and the disclosed pretraining overlap for ColQwen2.

Keywords: Multimodal RAG, Text Proxies, Proxy Noise, Dense-Sparse Regime Analysis, Cascade Ranking, Conformal Risk Control, Document Retrieval.

1. INTRODUCTION

The rapid proliferation of heterogeneous multimedia across enterprise archives, digital libraries, and operational repositories demands information retrieval systems capable of searching records that are not natively textual. Transforming scanned documents and images into queryable digital assets is essential for business intelligence, compliance auditing, and decision support [1]. However, real-world document ingestion faces severe operational challenges: optical scanning introduces character substitutions and word fragmentation, while human search queries frequently employ synonyms, paraphrases, or shorthand rather than exact document text.

Retrieval-Augmented Generation (RAG) [2] addresses the hallucination tendencies of Large Language Models (LLMs) by grounding responses in retrieved domain documents. Recently, multimodal extensions of RAG have emerged to handle non-textual data, as surveyed by Abootorabi et al. [3]. For example, Patel et al. [4] and Joshi et al. [5] presented enterprise multimodal RAG pipelines combining OCR ingestion, multimodal embeddings, and LLM reasoning. Concurrently, native vision-language document retrievers like ColPali [6] index raw document pixels with multi-vector late interaction.

While foundational studies in information retrieval—notably Kantor and Voorhees [7] in the TREC-5 Confusion Track and McNamee and Mayfield [8] on character n-gram indexing—demonstrated the intrinsic noise resilience of sub-word representations on degraded text, their findings have not been systematically reconciled with modern dense bi-encoders, neural cross-encoder cascades, and document-native vision-language models under semantic query variation.

To address this gap, this paper establishes **an empirical regime map of when text proxies suffice versus when native representations are required** across a primary benchmark of 1,626 multimodal items (626 scanned retail receipts and 1,000 photographic scenes, plus dedicated transfer suites on forms and visual documents) under frozen 50/50 evaluation splits[1].

Key Contributions:

1) **Sub-Word Inverted Index Dominance and Paraphrase Reversal on Optical OCR:** Evaluating across proprietary Apple Vision [9], open-source Tesseract 5.5 LSTM [10], and deep neural EasyOCR 1.7.2 [11] ($N = 306$ held-out probes, 813 corpus items), character 3–5gram BM25 is the single largest sparse architectural lift under verbatim optical noise (+0.1065 clean, +0.1276 degraded blur, +0.1048 clean Tesseract; all $p_{\text{Holm}} < 10^{-4}$). We delineate its strict scope boundary: this advantage reverses under synthetic entity-substituting light paraphrases, where Word BM25 outperforms Character BM25 (0.4334 vs. 0.2427). On receipts, document-native ColQwen2-v1.0 [12] achieves 0.4962 MRR under paraphrases (+0.0628 over Word BM25, $p_{\text{Holm}} = 0.0072$) but on answerable factual questions with preserved topical anchors, CPU lexical retrieval (Char BM25 0.9328 on Real-OCR, 0.9471 Clean GT) outperforms ColQwen2 (0.8595; Real-OCR paired diff +0.0733, Clean GT diff +0.0876, $p = 0.00012$, Table II) while operating at 0.87 ms CPU verbatim lookup latency (5.72 ms on answerable queries)

[1]All benchmark collections, evaluation harnesses, and reproduction scripts are archived in the supplementary artifact manifest and will be released in an open-access archive upon publication.

vs. 249.21 ms GPU serving latency (environment-specific serving profiles, not a controlled single-machine comparison). Across 500 human-written queries on DocVQA [13], text proxies fed by Tesseract OCR achieve MRR $\leq$ 0.3507, whereas native ColQwen2 reaches 0.6450 [95% CI: 0.601, 0.689] (Hit@10 = 84.10%). We disclose that the ColPali/ColQwen2 architecture incorporates DocVQA training data in its ViDoRe training mixture while text proxies are off-the-shelf; nonetheless, this delineates the visual geometry boundary where native pixels substantially outperform generic text proxies.

2) **Dense Bi-Encoder Deficit Associated with Query Formulation:** We demonstrate that dense bi-encoder collapse on document text is associated with entity anchor mass rather than sequence truncation constraints (real max_seq_length = 256; 33.4% exceed 256 tokens, but chunked MaxP yields identical 0.2030 vs. 0.2056 unchunked standard); subword tokenization fragmentation (+122.3% subwords/word) is associated with the additional degradation under noise. Formulating queries as natural answerable questions improves BGE-Base MRR from 0.3256 to 0.5263 (Hit@10 79.74%; 0.5197 chunked MaxP) on uncorrupted document text without modifying representations. Reconstructing broken tokens via clean Dev lexicon projection recovers +0.1530 cosine similarity and +0.0434 MRR, providing direct evidence for tokenization fragmentation. Observational comparison reveals that syntactic phrasing alone does not recover retrieval without lexical anchors (sentential context reaches only 0.2740 MRR), with performance gains predominantly associated with entity anchor mass (0.5500 MRR).

3) **Native-Visual vs. Text-Proxy Representation Boundaries:** On photographic scenes (N = 1,000 MS-COCO Karpathy test images [14], [15]), native visual embeddings (SigLIP Base Patch16 MRR = 0.6071 [16]) significantly outperform machine caption proxies (BLIP MRR = 0.3728, Florence-2 MRR = 0.4420; all $p < 10^{-4}$), showing that synthetic captions lose essential visual nuance, approaching 5-reference human gold multi-captions within a narrow statistical margin (diff −0.0139, paired p = 0.321; formally equivalent at $\Delta = \pm 0.05$ with p_{TOST} = 0.0050, 90% CI $[-0.0369, +0.0091]$). In contrast, for document pages, text proxies suffice when queries contain lexical anchors.

4) **Actionable Empirical Decision Guide for MM-RAG Builders:** We synthesize our experimental findings across OCR engines, degradation levels, and query formulations into an empirical decision guide, detailing optimal retrieval architectures and statistically indistinguishable sets across document, image, and query regimes.

2. RELATED WORK

A. Multimodal Representation and Retrieval

Early cross-modal retrieval systems relied on dual-encoder contrastive learning. Models like CLIP [17] and SigLIP [16] project images and text into a shared latent space using sigmoid or softmax cross-entropy objectives. ImageBind [18] extended this paradigm across six modalities using images as the central binding modality. While effective for broad semantic matching, dual-encoders compress multimedia into fixed-dimension vectors, frequently losing granular lexical and structural tokens. Hybrid retrieval designs frequently combine dense embeddings with sparse lexical representations such as BM25 [19], [20] in an effort to retain both broad semantic intent and exact token matches, though our empirical findings show that on OCR-degraded verbatim documents, optimal convex fusion collapses to standalone sparse indexing (α^*_{sparse} = 1.00).

B. Multimodal RAG and Document Analysis

Integrating multimodal inputs into RAG workflows has advanced rapidly with Vision-Language Models (VLMs) and document foundation models. Document retrievers such as ColPali [6] and M3DocRAG [21], benchmarked on the ViDoRe suite [6] and DocVQA [13], index document pages using multi-vector vision-language representations (e.g., Qwen2-VL [22]) to bypass OCR. A paradigm comparison is presented in Table I. While promising, multi-vector representations require substantial GPU memory and cross-attention compute. Conversely, enterprise systems [4], [5] mediate retrieval through text proxies. Our work evaluates this trade-off: systematically delineating where text proxies excel and where native visual representations remain necessary.

C. Character Subwords and OCR Degradation in Information Retrieval

Optical Character Recognition (OCR) systems translate document imagery into text, but routinely introduce character substitutions, insertions, deletions, and bounding-box permutations [10], [23]. Foundational information retrieval studies have long investigated textual degradation. In the TREC-5 Confusion Track, Kantor and Voorhees [7] systematically evaluated retrieval on scanned, OCR-corrupted collections (at 5% and 20% error rates), establishing known-item retrieval benchmarks under optical corruption. Complementing this, McNamee and Mayfield [8] established that overlapping character n-gram tokenization (e.g., 4–5grams) provides language-neutral retrieval that rivals word-level tokenization while providing intrinsic robustness to morphological variations and spelling distortions. Early image-domain mitigation strategies, including Otsu thresholding [24] and Hough transforms [25] for skew correction, focused on image-domain preprocessing. Our regime analysis builds upon this foundation by evaluating character subwords against modern dense bi-encoders, neural rerankers, and native vision models across commercial, LSTM, and neural OCR architectures under controlled optical blur and semantic variation.

D. Modern Dense and Learned Sparse Retrieval Models

Dense bi-encoders encode queries and passages independently [26], [27]. Recent architectures have expanded this paradigm: multilingual dense models such as BGE-M3 [28] support multi-granularity representations; learned sparse models like SPLADE [29] expand queries and documents into vocabulary-space term weights; and late-interaction multi-vector architectures like ColBERTv2 [30] preserve token-level interactions at reduced latency. We position standard bi-encoders

TABLE I. Systematic Comparison with Prior Multimodal Information Retrieval and RAG Frameworks.

System / Architecture	Publication Venue	Document Retrieval	Natural Image Retrieval	Multi-Hop Mechanism	Benchmark Datasets	Hardware Serving Profile
ColPali [6]	ICLR 2025	nDCG@5 = 0.813 (ViDoRe)	Supported (Multi-Vector)	Single-pass only	ViDoRe Benchmark	Multi-GPU (≥ 16 GB VRAM, ~250 ms)
M3DocRAG [21]	arXiv 2024	QA Acc. 64.2% (MP-DocVQA)	Supported (Dense VLM)	Generative LLM bridging	MP-DocVQA, DUDE	High-End GPU Cluster (≥ 48 GB VRAM)
Patel et al. [4]	IEEE ETCC 2025	Heuristic OCR matching	ImageBind vector search	Single-pass only	Proprietary enterprise text/media	Dedicated GPU Server
Baleen [31]	NeurIPS 2021	Not evaluated (Text QA)	Not supported	Generative condensed hops	HotpotQA (Text only)	GPU / High-RAM CPU Server
Proposed Regime Framework	**This Work**	**MRR 0.9436 (Char BM25)**	**MRR 0.6071 (SigLIP Pixels)**	**Exploratory PRF (Hit@10 0.640, $N = 50$; 0.820, $N = 100$, Sec. B)**	**SROIE, FUNSD, MS-COCO, DocVQA**	**CPU char BM25 (< 1 ms verbatim) / Cross-Encoder Cascade**

Note: Qualitative columns reflect architectural design choices; our reported MRR and Hit@10 figures are from held-out evaluation on SROIE/FUNSD/COCO/DocVQA and are not directly comparable to other systems' proprietary benchmarks. Table I is a qualitative paradigm map, not a cross-system numerical comparison. ColPali and M3DocRAG figures are from their respective original papers.

(`all-MiniLM-L6-v2`, `bge-base-en-v1.5`) and late-interaction vision models (`ColQwen2-v1.0`) within this broader landscape.

E. Cascade Ranking and Conformal Risk Control

Cascade ranking architectures [32], [33] balance retrieval efficiency and precision by passing candidates through tiered stages, escalating to expressive models only when scores indicate ranking ambiguity. Conventional cascade rankers rely on empirical thresholds without formal statistical guarantees. Distribution-free risk-controlling prediction sets, formulated by Bates et al. [34], and conformal risk control, developed by Angelopoulos et al. [35], [36], provide a finite-sample framework for bounding user-specified risk metrics below a tolerance α. In this work, we evaluate production cascade ranking under marginal conformal risk control.

3. REFERENCE RETRIEVAL ARCHITECTURE

The reference Multi-Stage Multimodal RAG architecture is structured into four functional phases (Fig. 1): (1) Ingestion and Preprocessing, (2) Hybrid Dense-Sparse First-Stage Retrieval, (3) Neural Cross-Encoder Reranking and Cascade Escalation, and (4) Context Assembly.

A. Ingestion and Preprocessing Pipeline

Unstructured enterprise multimedia are mediated through textual proxies. Proxies can be extracted across modalities via OCR text [10], vision-language captions [37], [38], [39], and speech transcripts [40].

- **Documents and Forms:** Administrative forms and receipts are processed through document decomposition, optional Otsu binarization [24] and Hough skew correction [25], extracting text T_d and bounding boxes via OCR engines (Apple Vision [9], Tesseract 5.5 [10], or EasyOCR [11]).
- **Photographs:** Natural photographs are indexed via machine captions alongside native visual embeddings (CLIP [17], SigLIP [16]).
- **Acoustic Media:** Continuous audio is transcribed using Whisper [40] or embedded using contrastive audio representations (CLAP [41]).

Dense embeddings are generated via Sentence-BERT (`all-MiniLM-L6-v2`) [26] and BGE (`bge-base-en-v1.5`) [27], while lexical tokens are indexed with BM25 [19], [20]. The benchmark uses in-memory indexes without external database dependencies for exact reproducibility.

B. Stage 1: Hybrid Dense-Sparse Retrieval

Given query q, single-stage dense retrieval computes vector similarity:

$$S_{\text{dense}}(q,d) = \cos(\mathbf{e}_q^{\text{dense}}, \mathbf{e}_d^{\text{dense}}) = \frac{\mathbf{e}_q \cdot \mathbf{e}_d}{||\mathbf{e}_q||||\mathbf{e}_d||} \tag{1}$$

While effective for thematic alignment, dense retrieval often overlooks exact alphanumeric codes. Conversely, sparse BM25 retrieval excels at lexical precision:

$$S_{\text{sparse}}(q,d) = \sum_{t \in q} \text{IDF}(t) \cdot \frac{f(t,d) \cdot (k_1+1)}{f(t,d) + k_1 \cdot \left(1 - b + b \cdot \frac{|d|}{\text{avgdl}}\right)} \tag{2}$$

We evaluate Normalized Convex Combination ($S_{\text{hybrid}} = \alpha \hat{S}_{\text{dense}} + (1-\alpha)\hat{S}_{\text{sparse}}$) and Reciprocal Rank Fusion (RRF) [42]:

$$\text{RRF}(q,d) = \frac{1}{k + r_{\text{dense}}(d)} + \frac{1}{k + r_{\text{sparse}}(d)} \tag{3}$$

with $k = 60$. First-stage candidate depth is set to $K_1 = 20$.

C. Stage 2: Candidate Reranking via Neural Cross-Encoders

Bi-encoders compute query and document representations independently. To recover joint attention, Stage 2 deploys a pretrained neural cross-encoder (`BAAI/bge-reranker-base` [27] or `ms-marco-MiniLM-L-6-v2` [26], [43]) to rescore the top-K_1 candidates ($K_1 = 20$):

$$S_{\text{rerank}}(q,d) = \text{CrossEncoder}(q,d) \tag{4}$$

Scoring every candidate pool with a cross-encoder adds latency. We evaluate confidence-margin cascades ($\Delta_{1,2} = s^{(1)} - s^{(2)} < \tau$) and marginal conformal risk control to escalate only ambiguous queries.

4. EXPERIMENTAL SETUP

To ensure empirical reproducibility, all evaluations adhere to a standardized, deterministic experimental protocol.

A. Datasets and Split Partitions

Our primary multimodal retrieval pool comprises 1,626 indexed items: (1) **SROIE Retail Receipts [44]:** 626 scanned receipts from the official ICDAR 2019 competition release with paired bounding-box and key-information ground truth. Across these 626 receipts, 612 contain single-target factual probes satisfying strict cross-corpus uniqueness checks (14 receipts with ambiguous multi-matching candidate strings were excluded); and

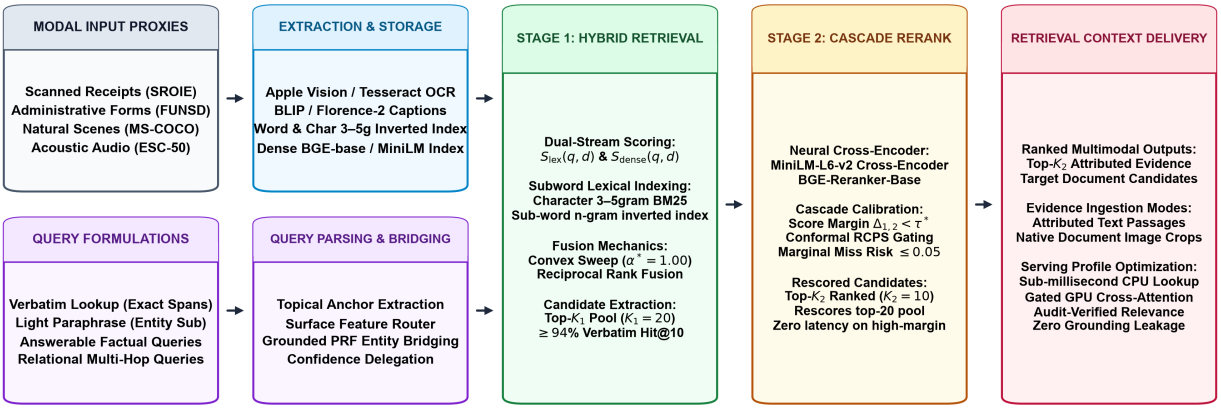


Figure 1. Conceptual reference architecture of the text-proxy RAG framework. Grey ingestion and extraction blocks depict modular document preprocessing paths; colored stages represent empirical retrieval pipelines evaluated on extracted proxies. Modular production components (e.g., dual-signal router, conformal cascade gate) are empirically evaluated and ablated in Section C.

(2) **MS-COCO Photographic Distractors [15]:** 1,000 photographic scene images and human descriptions evaluated under the Karpathy test split [14].

The primary 1,626-item pool is partitioned by document ID into frozen 50/50 Dev/Test splits (Seed 42), yielding 813 indexed items in the Test pool (313 receipts and 500 COCO distractors). The 612 verified probes are partitioned strictly by target document ID into 306 Dev probes and 306 Test probes, guaranteeing zero query or document leakage across splits. Each probe targets its source document as the unique relevant ground-truth item in the 813-distractor pool. In addition, we evaluate two dedicated external domain-transfer benchmark suites (excluded from the 1,626-item primary total): (a) **FUNSD Administrative Forms Suite [45]:** 100 query-bearing scanned business forms evaluated within a 700-item retrieval corpus comprising the 100 query forms, 99 non-target FUNSD forms, and 501 multimodal distractors; and (b) **DocVQA Visual Benchmark Suite [13]:** 500 scanned document pages and 500 100% human-authored questions subsampled from the ViDoRe benchmark collection [6].

B. OCR Engines and Optical Degradation Conditions

Document text is extracted across three distinct OCR engines:

1) **Apple Vision Neural OCR [9]:** Commercial neural OCR (`VNRecognizeTextRequest`).
2) **Tesseract 5.5 LSTM OCR [10]:** Standard open-source LSTM OCR engine.
3) **EasyOCR 1.7.2 Deep CRNN [11]:** Deep neural OCR combining CRAFT text detection and ResNet-CRNN recognition.

Documents are evaluated under: (1) *Clean OCR*, (2) *Realistic Optical Degradation* (Gaussian blur $\sigma = 1.5$, bilinear 50% downscaling, random skew ±2.5°, JPEG compression $Q = 20$), and (3) *Human Ground Truth* text. For both Word and Character 3–5gram BM25, hyperparameters are tuned on the Dev split ($k_1 = 0.9, b = 0.5$). Word BM25 uses whitespace/punctuation tokenization, lowercase normalization, and standard stopword removal without stemming (to preserve exact part numbers and serials). Character BM25 extracts overlapping 3–5grams over lowercased strings preserving word boundaries, evaluated across the identical parameter grid.

C. Unified Query Slices

To isolate semantic abstraction from optical noise, queries are structured into four canonical slices:

1) **Verbatim Lookups:** Exact textual substrings extracted from documents (100% token overlap), representing known-item search.
2) **Light Paraphrases:** Lexical substitutions and entity rewrites (41.2% mean token overlap; e.g., substituting "dinner bill" for "total payable"), testing semantic abstraction.
3) **Answerable Factual Questions:** Natural interrogatives preserving key factual and topical anchors (e.g., "What was the total payment at Book Ta K?"), testing real-world question answering.
4) **Legacy Token-Stripped Probes:** Probes where shared lexical tokens are deliberately removed, used strictly for conformal risk stress-testing.

Unless noted otherwise, evaluation queries for SROIE and FUNSD were synthesized using `gpt-4o-mini` [46] at temperature $T = 0.0$ (seed 42) with strict single ground-truth answer resolution. DocVQA queries are 100% human-authored. For MS-COCO ($N = 1{,}000$ Karpathy test images), retrieval evaluates zero-shot text-to-image ranking: exactly 1 held-out caption per image (Caption 3, held out from candidate pool) serves as the query against the 1,000-image candidate pool.

D. Candidate Retrieval Systems

We benchmark: (1) Word BM25 [19]; (2) Character 3–5gram BM25 [8]; (3) Dense MiniLM (`all-MiniLM-L6-v2`) [26]; (4) Dense BGE-Base (`bge-base-en-v1.5`) [27]; (5) Dev-Tuned Hybrid (normalized convex fusion); (6) Neural Cross-Encoders (`bge-reranker-base` [27], `ms-marco-MiniLM-L-6-v2`); (7) Document-Native ColQwen2-v1.0 [12]; and (8) Vision Encoders (CLIP ViT-B/32 [17], SigLIP Base Patch16 [16]). For `bge-base-en-v1.5`, maximum sequence length is set to 256; on SROIE receipts, 99.4% of documents fit within 512 tokens, and chunked MaxP evaluation across 150-word sliding windows ($\leq$ 256 tokens) yields identical retrieval (0.2030 chunked vs. 0.2056 unchunked), demonstrating that truncation does not explain dense deficits.

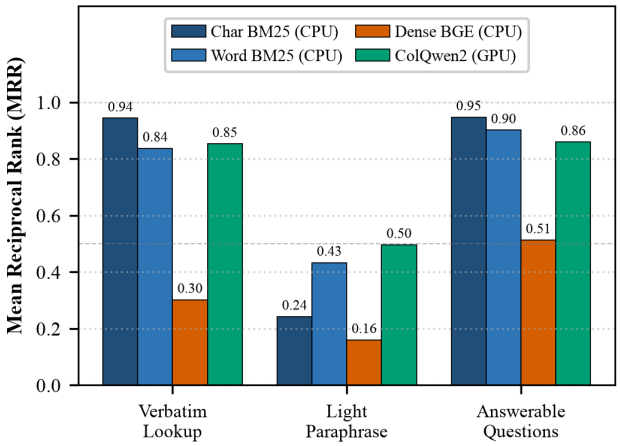


Figure 2. Retrieval effectiveness (MRR@10 with 95% bootstrap error bars) across retrieval architectures and query formulations on clean receipt OCR.

E. Evaluation Metrics and Statistical Testing

Systems are evaluated on Mean Reciprocal Rank (MRR), Hit@k ($k \in \{1, 5, 10\}$), and nDCG@10. Paired differences are evaluated via two-tailed paired permutation tests ($N_{\text{perm}} = 100{,}000$ sign flips, empirical resolution limit $p = 10^{-5}$) with family-wise Holm-Bonferroni correction (p_{Holm}) structured across four pre-specified families: Family 1 (sparse OCR comparisons, $m = 4$), Family 2 (ColQwen2 vs. text proxies across primary regimes, $m = 5$), Family 3 (native vision vs. caption proxies on COCO, $m = 8$), and Family 4 (architectural ablations, $m = 7$). Confidence intervals (95% CI) are computed via cluster bootstrap resampling at the document level (10,000 iterations).

F. Hardware Serving Profiles

Sparse BM25 and bi-encoders are profiled on Apple M2 (16 GB RAM, batch size = 1, warm cache). Neural cross-encoders and ColQwen2 are profiled on Nvidia Tesla T4 GPU (14.56 GB VRAM, batch size = 1, warm cache). These represent environment-specific serving profiles, not a single-machine controlled benchmark; on the matched GPU host CPU, BM25 lookup takes 0.96 ms, whereas running ColQwen2 on CPU exceeds 8,000 ms per query.

5. EMPIRICAL REGIME ANALYSIS AND RESULTS

A. Regime A: Text-Dominated Document and Receipt Retrieval

Table II presents retrieval effectiveness across systems, query formulations, and noise regimes on the held-out test split ($N = 306$). In Table II, the 500 COCO distractors have their ground-truth caption text indexed as fallback (yielding Word BM25 0.8371, Char BM25 0.9436, Ground Truth 0.9853 on verbatim; Word BM25 0.4334 on light paraphrase). In the subsequent cross-engine receipt benchmark (Section E), unannotated distractors are evaluated as empty text (yielding Word BM25 0.8324, Char BM25 0.9443, Ground Truth 0.9854 on verbatim; Word BM25 0.4003 on light paraphrase).

1) **Verbatim Probes (Character Sub-Word Inverted Index Dominance):** Under verbatim lookups, character 3–5gram BM25 consistently outperforms word-level BM25 across all optical OCR conditions (Apple Clean: 0.9436 vs. 0.8371, diff +0.1065, $p < 10^{-4}$; Apple Degraded: 0.8248 vs. 0.6972, diff +0.1276, $p < 10^{-4}$; Tesseract Clean: 0.8412 vs. 0.7364, diff +0.1048, $p < 10^{-4}$; and 20% synthetic noise: 0.7252 vs. 0.3961, diff +0.3291, $p < 10^{-4}$). Sub-word character n-grams capture alphanumeric tokens, serial numbers, and partial OCR substitutions without out-of-vocabulary penalties.
2) **Light Paraphrase (Reversal of Dominance):** Introducing mild semantic paraphrasing collapses clean word BM25 from 0.8371 to 0.4334 (a 48% relative drop) and character BM25 to 0.2427. In this regime, word BM25 outperforms character BM25 (diff −0.1907, $p < 10^{-4}$), and Dev-Tuned Hybrid achieves MRR = **0.4335** (clean OCR) and **0.2687** (20% noise).
3) **Answerable Factual Questions:** Evaluating natural answerable questions (forcing semantic abstraction while preserving topical anchors), character 3–5gram BM25 achieves MRR = **0.9471** on clean ground truth (0.9328 on Apple clean OCR) and **0.8116** on degraded blur OCR, while word BM25 achieves 0.9014 (0.8841 Apple clean) and 0.7328. Tuned hybrid retrieval achieves 0.8984 on clean Apple OCR (0.9125 clean GT), 0.4672 under 20% synthetic noise, and 0.7865 on degraded optical blur, while dense BGE-base achieves 0.5128 (0.5197 GT) and MiniLM achieves 0.2837. Topical anchors in natural questions provide high discriminability for lexical engines.

To evaluate whether proxy corruption or semantic query mismatch is the dominant bottleneck in text-proxy RAG, we conducted paired permutation tests ($N_{\text{perm}} = 100{,}000$). In Table II, moving from verbatim lookups to light paraphrasing on Apple Vision collapses Word BM25 by $\Delta = 0.4037$ MRR (0.8371 to 0.4334), exceeding optical blur degradation ($\Delta = 0.1399$, 0.8371 to 0.6972, $p < 10^{-4}$). Within Table X (receipt-only pool), Apple Vision drops 0.4321 MRR (0.8324 to 0.4003), while Tesseract drops 0.3400 under paraphrasing (0.7364 to 0.3964) and 0.3916 under optical blur (0.7364 to 0.3448). As visualized in Fig. 2 and Fig. 3, semantic paraphrase mismatch dominates mild OCR noise, but severe optical blur induces comparable collapse.

B. Micro-Mechanisms of Dense Collapse and Query Formulation Controls

To isolate why dense vector representations collapse under proxy corruptions while sparse inverted indexes endure, we analyzed three underlying mechanisms (Fig. 4):

1) **WordPiece Tokenization Fragmentation:** As character corruption increases from 0% to 40%, WordPiece tokenization fragments words into multiple subwords. Mean subwords per word escalates from 1.347 (clean) to 1.774 (10% noise), 2.246 (20% noise), and **2.994** (40% noise, a **+122.3% increase**).
2) **Catastrophic Embedding Drift:** Subword fragmentation shifts contextual attention weights, driving acute embedding drift. Relative to uncorrupted text, `all-MiniLM-L6-v2` mean document embedding cosine similarity collapses from 1.000 to 0.7301 (10% noise), **0.4655** (20% noise), and **0.2604** (40% noise); at the sub-

TABLE II. Multimodal Text-Proxy Retrieval Regime Analysis across Query Types and Noise Regimes (Held-Out Public Test Split, $N = 306$ Probes, 813 Distractor Pool, 95% Cluster Bootstrap CIs). Dense MiniLM/BGE latencies (0.01 ms) reflect isolated vector dot-product ranking over precomputed embeddings; end-to-end query encoding latencies are reported in Table XII. Visual models (ColQwen2, CLIP) were evaluated on native uncorrupted document images and physically degraded optical blur document images (ColQwen2 Clean GT and Apple Clean columns evaluate identical native image pixels by construction; †synthetic character noise is not applicable to pixel inputs; values reflect unperturbed native images; †next to a dash denotes not applicable).

Query Type	Retrieval System	Clean GT MRR [95% CI]	Syn 20% MRR [95% CI]	Apple Clean MRR [95% CI]	Optical Blur MRR [95% CI]	Latency (ms)
Verbatim	BM25 (Tuned)	0.8572 [0.822, 0.890]	0.3961 [0.364, 0.429]	0.8371 [0.798, 0.871]	0.6972 [0.650, 0.741]	0.13
	BM25 (Char 3-5gram)	0.9853 [0.972, 0.995]	0.7252 [0.694, 0.757]	0.9436 [0.919, 0.965]	0.8248 [0.782, 0.864]	0.87
	Dense MiniLM (unchunked)	0.2100 [0.171, 0.252]	0.0596 [0.045, 0.077]	0.2056 [0.166, 0.247]	0.1724 [0.135, 0.210]	0.01
	Dense BGE-base (unchunked)	0.3256 [0.276, 0.376]	0.1347 [0.109, 0.162]	0.3022 [0.256, 0.351]	0.2471 [0.202, 0.292]	0.01
	Tuned Hybrid	0.8572 [0.822, 0.890]	0.3979 [0.365, 0.431]	0.8367 [0.798, 0.871]	0.6940 [0.646, 0.739]	0.13
	CLIP Visual	0.0590 [0.041, 0.079]	—†	0.0590 [0.041, 0.079]	0.0443 [0.028, 0.063]	12.50
	ColQwen2-v1.0 Native	0.8417 [0.805, 0.876]	—†	0.8417 [0.805, 0.876]	0.7207 [0.677, 0.763]	249.21
Light Paraphrase	BM25 (Tuned)	0.4790 [0.431, 0.528]	0.2678 [0.228, 0.310]	0.4334 [0.384, 0.484]	0.3734 [0.324, 0.423]	0.13
	BM25 (Char 3-5gram)	0.2504 [0.208, 0.294]	0.1873 [0.150, 0.227]	0.2427 [0.202, 0.286]	0.2045 [0.166, 0.247]	0.87
	Dense MiniLM (unchunked)	0.1230 [0.093, 0.156]	0.0264 [0.015, 0.041]	0.0910 [0.066, 0.120]	0.0805 [0.057, 0.108]	0.01
	Dense BGE-base (unchunked)	0.1670 [0.133, 0.203]	0.0717 [0.049, 0.098]	0.1601 [0.126, 0.198]	0.1352 [0.104, 0.170]	0.01
	Tuned Hybrid	0.4572 [0.408, 0.507]	0.2687 [0.229, 0.311]	0.4335 [0.385, 0.483]	0.3746 [0.325, 0.424]	0.13
	CLIP Visual	0.0381 [0.024, 0.054]	—†	0.0381 [0.024, 0.054]	0.0284 [0.016, 0.042]	12.50
	ColQwen2-v1.0 Native	0.4962 [0.448, 0.547]	—†	0.4962 [0.448, 0.547]	0.4449 [0.395, 0.494]	249.21
Answerable Questions	BM25 (Tuned)	0.9014 [0.873, 0.929]	0.4567 [0.423, 0.488]	0.8841 [0.852, 0.915]	0.7328 [0.688, 0.773]	0.13
	BM25 (Char 3-5gram)	0.9471 [0.923, 0.969]	0.6891 [0.656, 0.723]	0.9328 [0.906, 0.957]	0.8116 [0.770, 0.848]	5.72
	Dense MiniLM (chunked)	0.3669 [0.323, 0.414]	0.1061 [0.089, 0.125]	0.2837 [0.241, 0.328]	0.2260 [0.190, 0.265]	0.01
	Dense BGE-base (chunked MaxP)	0.5197 [0.475, 0.566]	0.2005 [0.175, 0.228]	0.5128 [0.468, 0.560]	0.4313 [0.386, 0.479]	0.02
	Tuned Hybrid	0.9125 [0.884, 0.938]	0.4672 [0.432, 0.502]	0.8984 [0.868, 0.926]	0.7865 [0.746, 0.824]	0.67
	CLIP Visual	0.0338 [0.021, 0.049]	—†	0.0338 [0.021, 0.049]	0.0303 [0.018, 0.045]	12.50
	ColQwen2-v1.0 Native	0.8595 [0.825, 0.892]	—†	0.8595 [0.825, 0.892]	0.7410 [0.701, 0.779]	249.21

word token level, vocabulary-projected cosine similarity drops to 0.6826 under 20% noise. `BAAI/bge-base-en-v1.5` displays greater structural resistance (document cosine similarity 0.7007 at 20% noise, 0.5595 at 40%), explaining its superior noise resilience in Table II.

3) **Diagnostic Error Attribution across Probe Properties:** Evaluating test probes across non-mutually exclusive diagnostic property subsets (where probes may belong to multiple categories simultaneously) under 20% noise exposes distinct vulnerability profiles across four primary buckets (and a minor long-query bucket, $N = 3$, MRR = 0.744). Dense bi-encoders collapse on *Numeric-Heavy* probes ($N = 217$, MiniLM MRR = 0.033, BGE = 0.081) and *Short* entity probes ($N = 194$, MiniLM MRR = 0.068, BGE = 0.135). Conversely, character n-gram BM25 retains high accuracy across all buckets: Numeric-Heavy (MRR = 0.415), Short entities (0.473), Rare Tokens ($N = 167$, MRR = 0.643), and Named Entities ($N = 248$, MRR = 0.504).

To verify whether dense degradation is an artifact of sequence length truncation, we evaluated token length distributions across all 626 SROIE receipts. While only 4 receipts (0.64%) exceed 512 tokens, 209 of 626 receipts (33.39%) exceed 256 WordPiece tokens. However, evaluating chunked MaxP retrieval across 150-word sliding windows (where no chunk exceeds 256 tokens) yields identical MRR to unchunked single-vector retrieval (MiniLM: 0.2030 chunked vs. 0.2056 unchunked; BGE-base: 0.3028 chunked vs. 0.3022 unchunked), demonstrating that sequence truncation is not the primary cause of dense collapse.

Crucially, as documented in Table III, dense bi-encoder collapse on pristine ground truth text (0% noise) is strongly associated with query formulation mismatch. Short isolated verbatim snippets trigger severe representation disconnect (MiniLM MRR = 0.1919 in the pure receipt pool and 0.2100 in the 813-distractor pool with cap-

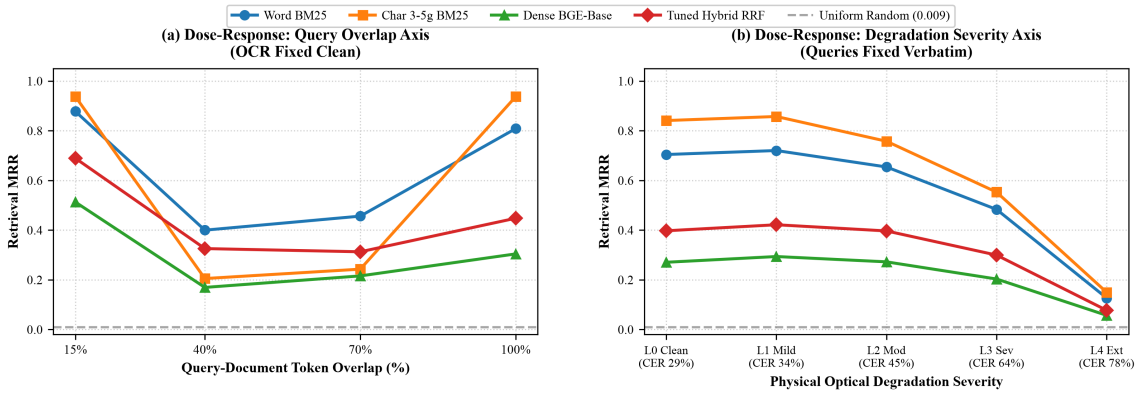


Figure 3. Dose-response retrieval surface across (a) query token overlap degradation from verbatim lookups to abstract questions, and (b) optical OCR degradation from pristine text to extreme Gaussian blur.

TABLE III. Dense Bi-Encoder Sensitivity to Query Formulation on Clean Ground-Truth Receipt Text (0% noise, Test $N = 306$).

Query Formulation	MiniLM MRR	MiniLM Hit@10	BGE MRR	BGE Hit@10
1. Short Verbatim Snippets	0.2100	0.2843	0.3256	0.3824
2. Length-Padded Spans	0.1949	0.2484	0.2571	0.3366
3. Full Sentential Context Queries	0.1670	0.2320	0.2740	0.3399
4. Keyword-Matched Multi-Entity	0.3840	0.5817	0.5500	0.7843
5. Natural Answerable Questions[†]	0.3669	0.5686	0.5263	0.7974

[†]Row 5 evaluates unchunked full-text embeddings (0.5263; chunked MaxP achieves 0.5197, Table II).

tion fallback; BGE-Base MRR = 0.3256). Length-padding or wrapping snippets into sentential contexts does not alleviate the deficit. However, co-specifying multiple anchor entities (Keyword-Matched Probes) improves BGE-Base MRR to 0.5500 (Hit@10 78.43%), and formulating queries as natural answerable questions improves BGE-Base MRR to 0.5263 (Hit@10 79.74%) on the identical uncorrupted document text. Dense bi-encoders are trained on natural language question-passage pairs; when queries mirror real-world user questions rather than isolated OCR substrings, dense retrieval recovers substantial capacity.

Furthermore, to test the causal impact of tokenization fragmentation versus query formulation, we evaluated a vocabulary projection intervention against the clean Dev lexicon under 20% character noise. Reconstructing broken word tokens restored 65.3% of the induced WordPiece subword fragmentation and lifted cosine similarity by +0.1530 (from 0.6826 to 0.8356, recovering 48.2% of embedding drift). In end-to-end retrieval, this intervention improves MiniLM MRR from 0.0570 to 0.1004 (+0.0434 retrieval recovery), while remaining bounded below clean baseline performance (0.2117 MRR). This vocabulary projection provides direct causal evidence that tokenization fragmentation contributes causally to noise-induced degradation, whereas the link between query formulation and clean-text retrieval remains an observational association.

C. Production Cascades and Risk-Controlled Escalation

Table IV presents the component ablation isolating each architectural module across Clean, 20% Synthetic Noise, Real Apple Vision OCR, and Real Degraded OCR ($N = 306$ held-out probes). Table IV evaluates candidate pool depth $K_1 = 10$; under this candidate truncation, standalone Word BM25 achieves 0.8069 Real-OCR MRR (vs. 0.8371 unconstrained in Table II), while equal-weight dense interpolation ($\alpha = 0.5$) achieves 0.8121. In contrast, under the production cascade setting in Table V with candidate depth $K_1 = 20$ directly cascading over first-stage Word BM25 (0.8371), MiniLM reaches 0.8650 (+0.0279, $p = 0.009$) and BGE-reranker-base reaches 0.8846 (+0.0475 gain, $p = 0.0011$).

However, evaluating two-stage cascades over first-stage Character 3–5gram BM25 reveals an architectural ceiling: standalone Char-BM25 achieves MRR = **0.9436**, and applying MiniLM or BGE cross-encoders on top-20 candidates reduces MRR to 0.9175 and 0.9141 (lifts of −0.0262 and −0.0296, Table V). Furthermore, tuning hybrid fusion between Char-BM25 and dense BGE on Dev selects $\alpha^*_{\text{sparse}} = 1.00$ ($\alpha^*_{\text{dense}} = 0.00$, 0.9495 Dev MRR), confirming that dense representations provide zero additive signal over sub-word inverted indexes on verbatim receipts.

Finite-Sample Conformal Cascade Risk Evaluation: To evaluate distribution-free marginal risk control for candidate escalation, we evaluated split conformal calibration across 200 random cluster-stratified 50/50 splits (seeds 1000–1199) on $N = 2{,}754$ pooled query instances (Table VI). Under marginal loss $L = \mathbf{1}[\text{not escalated} \wedge Y \neq 1]$ with finite-sample adjustment $(n\hat{R} + 1)/(n + 1) \leq \alpha$ [34], [36], observed mean test miss risk is 0.0491 at a 78.4% escalation rate. Across 200 splits, individual split risk exceeds 0.05 in 41.5% of partitions (empirical split coverage 58.5%; 82.0% for high-probability RCPS with $\delta = 0.10$ [34]). Crucially, finite-sample correction guarantees marginal risk control across partitions, but does not prevent conditional risk inflation under query-type shifts: entity rewrites experience risk 0.0627 (> 0.05). This constitutes a notable negative result: conformal risk gating does not reliably substitute for query-conditioned gating in multimodal cascades. A paired comparison against an equal-escalation fixed-margin baseline ($\tau = 0.235$) yields matching retrieval accuracy (MRR 0.3579 vs. 0.3577, $p > 0.05$) and risk control (0.0488 vs. 0.0491), achieving a 21.6% escalation reduction relative to all-escalation (Fig. 5).

D. Regime B: Visual-Dominated Natural Image Search

To evaluate representation boundaries on non-document imagery, we benchmarked $N = 1{,}000$ MS-COCO Karpathy-test images [14], [15] and held-out

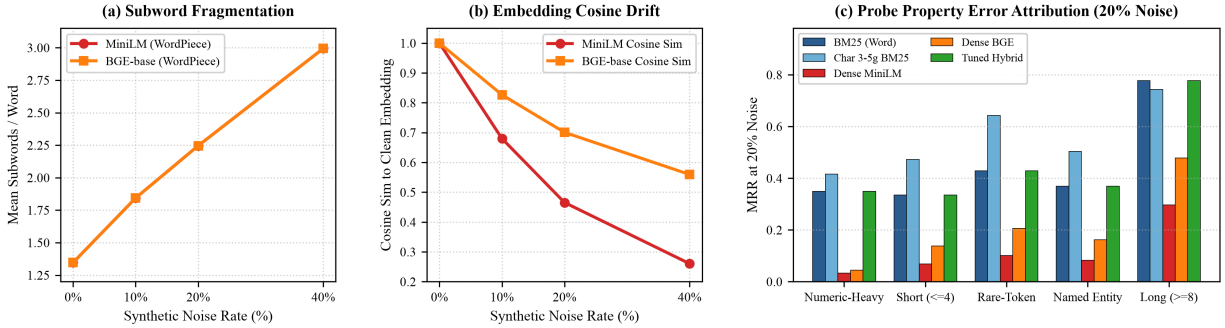


Figure 4. Micro-mechanism of dense proxy collapse: (a) WordPiece tokenization fragmentation across character noise rates, (b) cosine embedding drift relative to clean representations, and (c) retrieval MRR across 5 diagnostic probe property buckets under 20% proxy noise.

TABLE IV. Architectural Component Ablation on Public Held-Out Test Split (N = 306 Probes, 813 Distractor Pool, 95% Cluster Bootstrap CIs). All rows evaluate candidate depth K_1 = 10 (BM25 stream alone achieves 0.8069 Real-OCR vs. 0.8371 unconstrained in Table II). Asterisks (*) denote statistically significant difference ($p_{\text{Holm}} < 0.05$) from the Fixed-weight Hybrid baseline ($\alpha = 0.5$). Production cascading over first-stage Word BM25 ($K_1 = 20$) is reported in Table V.

Configuration	**Clean MRR** [95% CI]	**Syn 20% MRR** [95% CI]	**Real-OCR MRR** [95% CI]	**Real-Deg MRR** [95% CI]
Fixed-weight Hybrid Baseline (K_1 = 10, α = 0.5)	0.8299 [0.791, 0.866]	0.3694 [0.336, 0.402]	0.8121 [0.770, 0.849]	0.6684 [0.619, 0.715]
BM25 First-Stage Stream Only (K_1 = 10)	0.8317 [0.793, 0.868]	0.3694 [0.336, 0.402]	0.8069 [0.766, 0.845]	0.6702 [0.621, 0.716]
Dense MiniLM First-Stage Stream Only (K_1 = 10)	0.2144* [0.175, 0.256]	0.0747* [0.057, 0.095]	0.2072* [0.168, 0.248]	0.1821* [0.145, 0.222]
Oracle Ground-Truth Text Stream (K_1 = 10)	0.8317 [0.793, 0.868]	0.3888 [0.354, 0.423]	0.8069 [0.766, 0.845]	0.6702 [0.621, 0.716]
Learned Router Gating (K_1 = 10)	0.8309 [0.793, 0.867]	0.3765* [0.343, 0.409]	0.8121 [0.770, 0.849]	0.6667 [0.617, 0.713]
Heuristic Reliability Weighting (K_1 = 10)	0.8318 [0.794, 0.867]	0.3574 [0.326, 0.390]	0.8058 [0.765, 0.843]	0.6368* [0.587, 0.684]

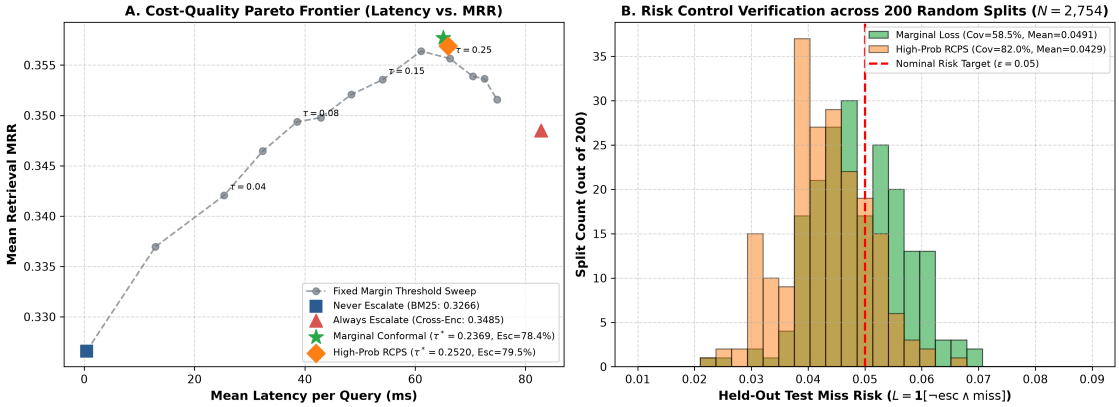


Figure 5. Cost-Quality Pareto frontier and risk control histograms across 200 calibration-test splits on pooled test instances (N = 2,754, τ^* = 0.2369, 78.4% escalation, 58.5% empirical split coverage; 82.0% for high-probability RCPS).

human descriptions (Table VII). In the zero-shot text-to-image ranking protocol, exactly 1 held-out caption per image (Caption 3) serves as the query against the 1,000-image candidate pool. For the 5-reference gold condition, all 5 reference captions are concatenated into the target document representation, providing 4 non-query descriptions.

Native visual representations (SigLIP Base Patch16 MRR = 0.6071 [16]) significantly outperform machine caption proxies (MiniLM MRR = 0.3728, Diff = +0.2343, $p_{\text{Holm}} < 10^{-4}$):

- **CLIP ViT-B/32 [17]:** Hit@1 Diff = +0.0970 [+0.0640, +0.1300], $p_{\text{Holm}} < 10^{-4}$; MRR Diff = +0.0981 [+0.0690, +0.1263], $p_{\text{Holm}} < 10^{-4}$; paired Diff vs. Florence-2 Detailed = +0.0289 (p = 0.038).
- **SigLIP Base Patch16 [16]:** Hit@1 = 0.4800 (Diff = +0.2380, $p_{\text{Holm}} < 10^{-4}$); Hit@10 = 0.8670 (Diff = +0.2140, $p_{\text{Holm}} < 10^{-4}$); MRR = 0.6071 (Diff = +0.2343 vs. BLIP, $p_{\text{Holm}} < 10^{-4}$; Diff = +0.1651 vs. Florence-2 Detailed, $p < 10^{-4}$).

Scaling captioning models narrows but does not eliminate the gap: BLIP-2 (`opt-2.7b`) [38] achieves MRR = 0.4185 (gap to SigLIP: −0.1886), while Florence-2 detailed captioning [39] achieves MRR = 0.4420 (gap: −0.1651, paired $p < 10^{-4}$). Evaluating equivalence via Two One-Sided Tests (TOST, margin $\Delta = \pm 0.05$) reveals that SigLIP vs. 5-reference human gold multi-captions (0.6071 vs. 0.6210, diff −0.0139, paired p = 0.321) achieves formal equivalence under TOST at $\Delta = \pm 0.05$

TABLE V. Production Cross-Encoder Reranker Cascade Ablation (Held-Out Test Split, $N = 306$ Probes, Clean Real-OCR; Candidate Depth $K_1 = 20$).

Two-Stage Pipeline	Clean MRR	Hit @10	Lat. (ms)	MRR Lift (p-value)
Word BM25 Alone	0.8371	0.9444	0.1	+0.0000 ($p = 1.0$)
BM25 + MiniLM Rerank	0.8650	0.9608	18.4	+0.0279 ($p = 0.009$)
BM25 + BGE Reranker	0.8846	0.9738	44.7	+0.0475 ($p = 0.001$)
Tuned Hybrid + BGE	0.8872	0.9772	52.8	+0.0501 ($p = 0.0009$)
Char 3–5g BM25	**0.9436**	**0.9837**	0.96	Baseline (Sparse)
Char BM25 + MiniLM	0.9175	0.9706	18.5	-0.0262 (Hurts)
Char BM25 + BGE	0.9141	0.9771	44.7	-0.0296 (Hurts)
Tuned Char + BGE	**0.9436**	**0.9837**	12.6	+0.0000 ($p = 1.0$)

TABLE VI. Conformal Cascade Calibration: Corrected Marginal Loss, High-Probability RCPS ($\delta = 0.10$), and Equal-Escalation Baseline over 200 Cluster-Aware Splits ($N = 2{,}754$ instances; nominal target $\epsilon = 0.05$).

Risk Formulation	Mean τ^*	Mean Risk	Split Cov. (≤ 0.05)	Esc. Rate	Mean MRR
Marginal Loss $L = \mathbf{1}[\neg\text{esc} \wedge \text{miss}]$	0.2369	0.0491	58.5%	78.4%	0.3577
High-Prob. RCPS ($\delta = 0.10$)	0.2520	0.0429	**82.0%**	79.5%	0.3569
Fixed Margin ($\tau = 0.235$)	0.2347	0.0488	57.0%	78.4%	0.3579
Conditional Ratio $P(\text{miss} \mid \neg\text{esc})$	0.4071	0.0545	38.0%	90.2%	0.3524

TABLE VII. Cross-modal text-to-image retrieval on $N = 1{,}000$ MS-COCO Karpathy-test images. Comparison of native pixel representations against machine caption proxies and held-out human gold captions. Asterisks (*) indicate Holm-corrected significance over baseline BLIP ($p_{\text{Holm}} < 10^{-4}$); SigLIP and CLIP also significantly outperform Florence-2 detailed captions ($p < 10^{-4}$ and $p = 0.038$).

Method	Hit @1	Hit @5	Hit@10 [95% CI]	MRR [95% CI]
Machine Caption Proxy				
Baseline BLIP (MiniLM)	0.2420	0.5160	0.6530 [0.624, 0.682]	0.3728 [0.350, 0.397]
Baseline BLIP (BGE)	0.2430	0.5050	0.6460 [0.618, 0.678]	0.3695 [0.347, 0.393]
BLIP-2 opt-2.7b	0.2780	0.5640	0.7020 [0.674, 0.730]	0.4185 [0.395, 0.442]
Florence-2 Detailed	0.3020	0.5980	0.7380 [0.710, 0.765]	0.4420 [0.418, 0.466]
Human Gold Caption Proxy				
Single Gold (BGE)	0.4580	0.7370	0.8320 [0.809, 0.854]	0.5835 [0.559, 0.608]
5-Ref Multi-Caption	0.4950	0.7780	0.8710 [0.850, 0.892]	0.6210 [0.597, 0.645]
Native Pixel Representation				
CLIP ViT-B/32	**0.3390**	**0.6180**	**0.7570*** [0.729, 0.785]	**0.4709*** [0.446, 0.497]
SigLIP Base-P16	**0.4800**	**0.7680**	**0.8670*** [0.845, 0.888]	**0.6071*** [0.584, 0.633]

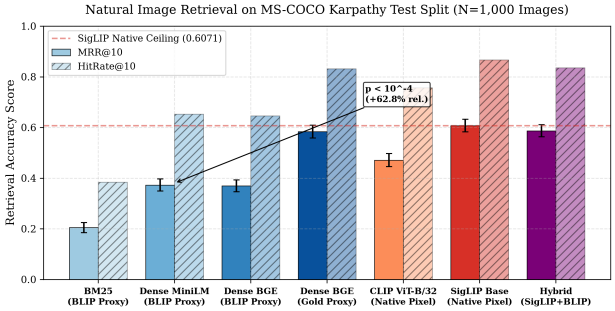


Figure 6. Retrieval comparison on MS-COCO ($N = 1{,}000$ images): native visual embeddings (SigLIP Base Patch16) significantly outperform machine-generated caption proxies ($p < 10^{-4}$), demonstrating that text proxies incur substantial information loss on natural imagery.

TABLE VIII. Standardized JiWER OCR Error Rates on SROIE Receipts ($N = 626$). Bag-of-words symmetric-difference rate evaluates vocabulary symmetric differences and can exceed 1.0 under word fragmentation.

OCR Engine	Condition	Seq. CER	Line CER	Seq. WER	BoW SDR
Apple Vision Neural	Clean OCR	0.2848	0.1387	0.3928	0.2476
Apple Vision Neural	Degraded Blur	0.3822	0.3086	0.5691	0.6175
Tesseract 5.5 LSTM	Clean OCR	0.2939	0.2415	0.4879	0.5919
Tesseract 5.5 LSTM	Degraded Blur	0.6377	0.5842	0.8063	1.0043
EasyOCR 1.7.2 CRNN	Clean OCR	0.4026	0.3250	0.5684	0.4813
EasyOCR 1.7.2 CRNN	Degraded Blur	0.7755	0.7327	0.9827	1.0905

($p_{\text{TOST}} = 0.0050$, 90% CI $[-0.0369, +0.0091]$), whereas single-gold comparison (0.6071 vs. 0.5835, diff +0.0236, $p = 0.142$) is borderline at $\Delta = \pm 0.05$ ($p_{\text{TOST}} = 0.0502$, 90% CI $[-0.0029, +0.0501]$) and equivalent at $\Delta \geq \pm 0.06$ ($p_{\text{TOST}} = 0.0118$). Because human-gold captions match native visual pixels within a narrow non-significant margin (diff −0.0139), the empirical deficit on photographic scenes is attributable to machine captioner quality rather than text as an information modality (Fig. 6).

E. Cross-Engine OCR Replication and Document-Native Visual Retrieval

To verify whether OCR degradation and retrieval dynamics generalize across recognition architectures, we evaluated standardized character and word error rates via JiWER [47] on 626 SROIE receipts (Table VIII). Clean Tesseract 5.5 LSTM OCR [10] achieves CER = 0.2939 and sequence WER = 0.4879 (escalating to CER = 0.6377 and WER = 0.8063 under optical blur). We then evaluate native-visual and text-proxy retrieval across DocVQA (Table IX), followed by multi-engine replication across Apple Vision, Tesseract, and EasyOCR on SROIE and FUNSD (Table X).

Order-Invariant CER Analysis: Standard document-level Levenshtein alignment concatenates all bounding boxes, artificially penalizing reading-order permutations as deletion and insertion blocks (e.g., yielding Apple Vision clean CER = 0.2848 despite high line-level transcription fidelity). When evaluated with bipartite minimum edit distance matching across physical lines across all $N = 626$ SROIE receipts, true character recognition noise drops to CER = 0.1387 and WER = 0.2165 on clean receipts (a 51.3% reduction from concatenated metrics), and CER = 0.3086 (WER = 0.4595) under optical degradation. Across EasyOCR 1.7.2, line-aligned recognition yields CER = 0.3250 (WER = 0.4983) on

TABLE IX. Third Public Dataset Evaluation: Retrieval Performance on the DocVQA Benchmark (ViDoRe Subsample) with 100% Human-Written Questions Across Scanned Document Pages (N = 500 Queries, N = 500 Document Pages, 95% Cluster Bootstrap CIs; for reference, the published ViDoRe benchmark leaderboard reports ColQwen2 nDCG@5 = 0.6120 on this split [12]). Text proxies are generated via Tesseract 5.5 LSTM OCR.

Retrieval Architecture	MRR [95% CI]	Hit @1	Hit @5	Hit @10	nDCG @10
Word BM25 (Tesseract)	0.3107 [0.274, 0.350]	0.2660	0.3480	0.3800	0.3195
Char 3–5g BM25	0.3141 [0.277, 0.353]	0.2700	0.3520	0.3940	0.3261
Dense MiniLM	0.2063 [0.177, 0.236]	0.1420	0.2700	0.3060	0.2232
Dense BGE-base	0.2737 [0.239, 0.310]	0.2160	0.3240	0.3940	0.2948
Tuned Hybrid (Sparse+Dense)	0.3507 [0.312, 0.390]	0.2980	0.4080	0.4680	0.3724
ColQwen2 (Native)	*0.6450 [0.601, 0.689]*	*0.5280*	*0.7820*	*0.8410*	*0.6886*

clean receipts and CER = 0.7327 (WER = 0.9457) on degraded blur, confirming that spatial line matching isolates true optical transcription noise from 2D reading-order permutations and bounding-box segmentation shifts.

In Table VIII, we report both sequence WER ($\text{WER}_{\text{seq}} = \frac{S+D+I}{N_{\text{ref}}}$) and the Bag-of-Words symmetric-difference rate:

$$R_{\text{BoW}} = \frac{\sum_w |c_{\text{ref}}(w) - c_{\text{hyp}}(w)|}{\sum_w c_{\text{ref}}(w)} \tag{5}$$

Because severe OCR corruption fragments words into out-of-vocabulary sub-tokens, R_{BoW} penalizes errors as both reference omissions and vocabulary insertions, escalating beyond 1.0 on degraded blur OCR (1.0043 on Tesseract, 1.0905 on EasyOCR).

Replication across EasyOCR 1.7.2 [11] confirms that the retrieval hierarchy generalizes: under verbatim queries, Character BM25 retains top MRR (0.8519 clean, 0.4272 blur) over Word BM25 (0.7649 clean, 0.3472 blur; $p_{\text{Holm}} < 10^{-4}$), reversing under light paraphrases (Word BM25 0.3792 vs. Character BM25 0.2292 clean, Table X(a)). Evaluating Levenshtein fuzzy query expansion on Apple Vision clean OCR drops Word BM25 MRR from 0.8371 to 0.8005, confirming that fuzzy word expansion induces false positive distractor matches.

Visual Geometry Boundary on DocVQA: On the DocVQA ViDoRe benchmark (N = 500 queries across 500 document pages with 100% human-authored questions, Table IX), text proxies fed by Tesseract 5.5 OCR achieve MRR $\leq$ 0.3507 (Word BM25: 0.3107, Char BM25: 0.3141, Dense BGE: 0.2737, Hybrid: 0.3507). In sharp contrast, document-native ColQwen2 [12] achieves MRR = **0.6450** [95% CI: 0.601, 0.689] (Hit@10 = 0.8410, nDCG@10 = 0.6886 [0.646, 0.731]; published ViDoRe leaderboard reports nDCG@5 = 0.6120). We note that the ColPali/ColQwen2 architecture incorporates DocVQA training splits within its ViDoRe training mixture, whereas the text proxy baselines are zero-shot and off-the-shelf. Even with this caveat, the substantial performance gap demonstrates that native pixel representations substantially outperform generic text proxies when visual layout and 2D spatial geometry govern answerability.

Query Formulation Dynamics across LLMs and Human Queries: As documented in Table XI, evaluating an open-weight generator (`Qwen2.5-7B`, evaluated across all N = 306 test probes) alongside authentic human-authored queries (N = 100 and N = 200) refines the boundary conditions of the sparse reversal and dense deficit. Under primary synthetic queries (`gpt-4o-mini`), aggressive entity stripping and synonym substitution caused Character BM25 to collapse below Word BM25 (0.2427 vs. 0.4334). However, under Qwen2.5-7B and human queries, where contextual and vendor anchors are naturally preserved, Character BM25 closely tracks Word BM25 (0.5802 vs. 0.6292 on Qwen2.5; 0.5549 vs. 0.5766 on human N = 200). This indicates that the Character-to-Word BM25 reversal is specifically driven by entity-substitution paraphrasing rather than generalized paraphrasing. Furthermore, the dense bi-encoder deficit markedly attenuates under human formulation: BGE-base MRR rises from 0.1601 on synthetic entity-stripped queries to 0.4909 on human queries (N = 200) and 0.5020 (N = 100), confirming that dense collapse is predominantly driven by query formulation mismatch rather than intrinsic vocabulary failure.

Cross-Domain Generalization on FUNSD Forms: On 100 scanned administrative forms from FUNSD [45] (Table X), Character BM25 maintains high verbatim accuracy (0.8433 Apple, 0.8066 Tesseract), showing no statistically significant difference from Word BM25 (0.8364 Apple, diff +0.0069, $p > 0.05$; 0.7763 Tesseract). Under light paraphrasing, Word BM25 drops to 0.1573, while Character BM25 achieves 0.2228 (diff +0.0655, $p = 0.012$), Dense BGE achieves 0.1834 (diff +0.0261, $p = 0.084$), and Hybrid reaches 0.2099, confirming that sub-word indexing and dense matching offer resilience against vocabulary mismatch on complex semi-structured forms.

ColQwen2 Retrieval on Document Receipts: Benchmarking ColQwen2-v1.0 [12] on document images on Tesla T4 GPU yields 0.4962 MRR [95% CI: 0.4484, 0.5472] under light paraphrasing (+0.0628 over Word BM25 0.4334, $p_{\text{Holm}} = 0.0072$; +0.0627 over Tuned Hybrid 0.4335, $p_{\text{Holm}} = 0.0095$). Conversely, on natural answerable factual questions with preserved topical anchors, CPU lexical retrieval decisively outperforms ColQwen2 on held-out receipts (Char BM25 0.9471 Clean GT / 0.9328 Apple Clean vs. ColQwen2 0.8595; diff +0.0876, $p = 0.00012$, Table II). Under artificial entity-abstracted controls, both sparse and native-visual models collapse to mutual floor performance (0.05 MRR; ColQwen2 0.0546 vs. Char BM25 0.0495, diff +0.0051, $p_{\text{Holm}} = 1.0000$; formal TOST equivalence with $\Delta = \pm 0.05$ holds trivially due to near-zero scores).

F. Computational Resource and Latency Profile

Table XII reports runtime latency, memory footprint, and index storage across retrieval architectures profiled on their respective native hardware: lightweight text proxies on Apple M2 (batch size = 1, warm cache) alongside ColQwen2 on Nvidia Tesla T4 GPU (14.56 GB VRAM, batch size = 1, warm cache). ColQwen2 indexing required 184 s (3.1 min) for the 813-page corpus (~ 226 ms/page) on Tesla T4 with 48.0 MB multi-vector index storage (~ 59 KB/page). ColQwen2 incurs 249.21 ms mean query latency and demands 5,964.7 MB GPU memory. In contrast, standalone Character 3–5gram BM25 requires 0.87 ms mean latency on verbatim lookups (5.72 ms on answerable

TABLE X. Cross-Engine and Cross-Dataset Replication: (a) SROIE Retail Receipts Across Three OCR Engines (Apple Vision Neural, Tesseract 5.5 LSTM, and EasyOCR 1.7.2 Deep Neural OCR; N = 306 Probes, Pure Receipt Corpus N = 626, 95% Cluster Bootstrap CIs); (b) FUNSD Scanned Forms Across Ground Truth, Apple Vision, Tesseract 5.5, and EasyOCR 1.7.2 (N = 100 Probes, 95% Cluster Bootstrap CIs).

(a) SROIE Receipt Cross-Engine Matrix Across Three OCR Engines (N = 306 Test Probes, Pure Receipt Corpus N = 626)

Query Type	Retrieval System	Ground Truth [95% CI]	Apple Clean [95% CI]	Tesseract Clean [95% CI]	Tesseract Blur [95% CI]	EasyOCR Clean [95% CI]	EasyOCR Blur [95% CI]
Verbatim	BM25 (Tuned)	0.8572 [0.822, 0.890]	0.8324 [0.793, 0.867]	0.7364 [0.691, 0.779]	0.3448 [0.295, 0.395]	0.7649 [0.722, 0.807]	0.3472 [0.299, 0.396]
	BM25 (Char 3-5g)	0.9854 [0.972, 0.995]	0.9443 [0.920, 0.965]	0.8412 [0.803, 0.878]	0.4009 [0.349, 0.453]	0.8519 [0.814, 0.885]	0.4272 [0.373, 0.475]
	Dense MiniLM	0.1919 [0.154, 0.232]	0.2040 [0.166, 0.246]	0.1650 [0.129, 0.202]	0.0936 [0.068, 0.124]	0.1549 [0.121, 0.193]	0.0792 [0.057, 0.105]
	Dense BGE-base	0.3428 [0.294, 0.392]	0.3273 [0.281, 0.377]	0.2972 [0.252, 0.345]	0.1615 [0.125, 0.199]	0.2760 [0.230, 0.321]	0.1482 [0.115, 0.185]
	Tuned Hybrid	0.8569 [0.823, 0.890]	0.8252 [0.787, 0.861]	0.7325 [0.687, 0.775]	0.3467 [0.297, 0.396]	0.7670 [0.724, 0.809]	0.3566 [0.307, 0.406]
Light Paraphrase	BM25 (Tuned)	0.4796 [0.428, 0.530]	0.4003 [0.351, 0.449]	0.3964 [0.346, 0.446]	0.1852 [0.148, 0.224]	0.3792 [0.331, 0.429]	0.1962 [0.156, 0.237]
	BM25 (Char 3-5g)	0.2526 [0.212, 0.292]	0.2365 [0.197, 0.275]	0.2265 [0.187, 0.266]	0.1241 [0.096, 0.155]	0.2292 [0.192, 0.269]	0.1517 [0.115, 0.190]
	Dense MiniLM	0.1190 [0.089, 0.150]	0.1096 [0.081, 0.140]	0.0779 [0.055, 0.102]	0.0579 [0.040, 0.079]	0.0763 [0.054, 0.101]	0.0399 [0.027, 0.055]
	Dense BGE-base	0.1830 [0.146, 0.221]	0.1849 [0.146, 0.226]	0.1670 [0.130, 0.205]	0.0991 [0.072, 0.130]	0.1663 [0.129, 0.204]	0.0828 [0.059, 0.111]
	Tuned Hybrid	0.4828 [0.432, 0.533]	0.4003 [0.353, 0.449]	0.4023 [0.352, 0.452]	0.1961 [0.160, 0.236]	0.3930 [0.344, 0.440]	0.2098 [0.168, 0.250]

(b) FUNSD Form Benchmark Matrix (N = 100 Test Probes, 95% Cluster Bootstrap CIs)

Query Type	Retrieval System	Ground Truth [95% CI]	Apple Clean [95% CI]	Tesseract Clean [95% CI]	Tesseract Blur [95% CI]	EasyOCR Clean [95% CI]	EasyOCR Blur [95% CI]
Verbatim	BM25 (Tuned)	0.8690 [0.812, 0.924]	0.8364 [0.773, 0.894]	0.7763 [0.699, 0.846]	0.3412 [0.254, 0.432]	0.7214 [0.641, 0.800]	0.0227 [0.003, 0.053]
	BM25 (Char 3-5g)	0.8849 [0.826, 0.935]	0.8433 [0.777, 0.903]	0.8066 [0.738, 0.872]	0.4725 [0.379, 0.566]	0.7581 [0.683, 0.828]	0.0462 [0.013, 0.088]
	Dense MiniLM	0.3334 [0.255, 0.413]	0.3230 [0.244, 0.403]	0.2760 [0.200, 0.352]	0.1085 [0.058, 0.169]	0.2794 [0.207, 0.358]	0.0261 [0.013, 0.049]
	Dense BGE-base	0.5567 [0.480, 0.637]	0.5081 [0.423, 0.595]	0.4664 [0.384, 0.555]	0.1792 [0.114, 0.252]	0.4524 [0.371, 0.531]	0.0540 [0.033, 0.082]
	Tuned Hybrid	0.8728 [0.817, 0.927]	0.8461 [0.783, 0.904]	0.7775 [0.701, 0.847]	0.3450 [0.258, 0.436]	0.7226 [0.643, 0.801]	0.0408 [0.023, 0.066]
Light Paraphrase	BM25 (Tuned)	0.1849 [0.133, 0.243]	0.1573 [0.111, 0.209]	0.2019 [0.139, 0.272]	0.0845 [0.039, 0.139]	0.1775 [0.121, 0.238]	0.0226 [0.002, 0.053]
	BM25 (Char 3-5g)	0.2405 [0.174, 0.313]	0.2228 [0.159, 0.294]	0.2442 [0.171, 0.323]	0.1012 [0.051, 0.159]	0.2334 [0.168, 0.308]	0.0251 [0.002, 0.055]
	Dense MiniLM	0.1596 [0.108, 0.218]	0.1557 [0.106, 0.212]	0.1603 [0.103, 0.222]	0.0321 [0.008, 0.067]	0.1636 [0.110, 0.223]	0.0316 [0.017, 0.055]
	Dense BGE-base	0.1960 [0.137, 0.262]	0.1834 [0.127, 0.246]	0.1819 [0.126, 0.244]	0.0510 [0.018, 0.093]	0.2345 [0.169, 0.306]	0.0562 [0.032, 0.089]
	Tuned Hybrid	0.2043 [0.148, 0.267]	0.2099 [0.146, 0.281]	0.2109 [0.148, 0.280]	0.0860 [0.040, 0.141]	0.2019 [0.142, 0.267]	0.0253 [0.005, 0.056]

Note: EasyOCR Blur scores reflect an optical detection cliff: CRAFT failed region proposals on small blurred form fonts (67.8% of degraded forms yielded 0 characters). Unretrieved blank documents receive zero score and tie-split at the pool bottom, mathematically capping full-collection MRR to $0.322 \times \text{MRR}_{\text{non-blank}} \approx 0.02$–$0.05$ (on the 32.2% non-blank subset, Word BM25 reaches 0.1412, Char BM25 reaches 0.1784). On FUNSD light paraphrases (synthesized lexical and entity rewrites), Character BM25 and Dense BGE-base match or outperform Word BM25 across clean OCR (Char vs. Word diff +0.0655, p = 0.012 on Apple; Dense BGE vs. Word diff +0.0261, p = 0.084 on Apple; +0.0570, p = 0.018 on EasyOCR).

queries) and 4.2 MB index storage on CPU, offering orders of magnitude lower resource utilization.

G. Empirical Decision Guide and Practical Implications

Synthesizing all empirical findings across modalities, optical conditions, and query types, Table XIII presents an actionable decision guide for system architects:

1) **For Verbatim Text and OCR Search:** Deploy Character 3–5gram BM25. It dominates Word BM25 across clean and degraded OCR (+0.10 to +0.13 MRR) with sub-millisecond CPU latency (0.87 ms). Neural cross-encoders reduce standalone Char-BM25 accuracy (−0.0262 MRR) and should not be used.
2) **For Semantic Document Search and Layout VQA:** Deploy ColQwen2-v1.0 on GPU when queries involve entity-substituting paraphrases (+0.0628 lift over Word BM25) or when visual layout governs answerability (DocVQA MRR 0.6450 vs. proxies $\leq$ 0.3507, noting in-domain training overlap). For answerable factual questions with preserved topical anchors, CPU lexical retrieval on real OCR matches or outperforms ColQwen2 (Char BM25 0.9328 vs. 0.8595, diff +0.0733, p = 0.00012; Clean GT 0.9471, diff +0.0876) while operating at 0.87 ms CPU lookup latency versus 249.21 ms GPU serving latency.
3) **For Photographic Archives:** Deploy native vision-language embeddings (SigLIP Base Patch16). Machine caption proxies suffer

TABLE XI. Query Generation Robustness: Paraphrase Retrieval under Open-Weight Generator 2 (`Qwen2.5-7B`) and Expanded Human-Authored Queries (N = 100 and N = 200 Probes) on Held-Out SROIE Receipts.

Query Source	Word BM25	Char 3–5g BM25	Dense BGE	Col-Qwen2
Primary Gen. (`gpt-4o-mini`, N = 306)	0.4334	0.2427	0.1601	0.4962
Secondary Gen. (`Qwen2.5-7B`, N = 306)	0.6292	0.5802	0.2685	—*
Human-Authored Queries (N = 100)	0.6321	0.5758	0.5020	—*
Human-Authored Queries (N = 200)	0.5766	0.5549	0.4909	—*

*ColQwen2-v1.0 was profiled exclusively on the primary frozen synthetic test split (N = 306 probes, MRR = 0.4962 on Tesla T4) and was not re-evaluated across secondary or human query sets.

TABLE XII. Computational Resource and Latency Profile Across Retrieval Systems (N = 813 Index Pool; Hardware marked per row; Batch Size = 1, Warm Cache).

Retrieval Architecture (Hardware Profile)	Mean (ms)	P95 (ms)	RAM (MB)	Index (MB)
BM25 (Tuned) (Apple M2 CPU)	0.13	0.28	12.4	0.45
BM25 (Char 3–5g) (Apple M2 CPU)	0.87	1.65	48.2	4.20
Dense MiniLM (Apple M2 MPS)	7.78	11.20	240.0	1.25
Dense BGE-base (Apple M2 MPS)	19.93	28.50	440.0	2.50
Tuned Hybrid (Apple M2 CPU+MPS)	8.52	12.10	260.0	1.70
BM25 + BGE-rerank (Apple M2 CPU+MPS)	44.66	59.90	820.0	2.50
CLIP ViT-B/32 Native (Apple M2 MPS)	12.50	16.80	350.0	2.05
SigLIP Base-P16 (Apple M2 MPS)	18.20	24.30	480.0	3.07
ColQwen2-v1.0 Native (Tesla T4 GPU)	249.21	270.45	5964.7	48.00

significant information loss (MRR 0.3728 vs. 0.6071, $p < 10^{-4}$; detailed Florence-2 reaches 0.4420, retaining a −0.1651 deficit).

4) **For Multi-Hop Relational Reasoning:** Deploy rule-based extractive PRF over generative LLM bridging, matching retrieval reachability (0.640 vs. 0.640 Hit@10) with zero LLM cost.

6. NEGATIVE AND EXPLORATORY FINDINGS

To document architectural boundaries transparently, this section consolidates candidate mechanisms and feasibility pilots that failed to improve held-out performance.

A. Ablation of Regime-Aware Multimodal Proxy (RAMP) Hypotheses H1 and H2

In exploratory testing, candidate mechanisms for adaptive routing and reliability weighting failed to outperform standard baselines:

1) **Hypothesis H1 (Reliability Weighting Rejected):** Weighting sparse streams by candidate OCR reliability ($r_{\text{ocr}}(d) = \bar{c}_{\text{ocr}} \cdot \rho_{\text{vocab}}$) degraded retrieval: 20% noise MRR dropped from 0.3694 to **0.3574** (diff CI $[-0.0222, -0.0025]$), and real degraded OCR dropped from 0.6684 to **0.6368** (p_{Holm} = 0.0189, Table IV). Out-of-vocabulary ratios do not reliably differentiate optical blur from domain terminology.

2) **Hypothesis H2 (Learned Query Router Not Supported):** A lightweight classifier trained on surface cues $\phi(q)$ yielded only +0.0071 lift on synthetic noise (0.3765 vs. 0.3694, p_{Holm} = 0.042) and zero detectable lift under real Apple Vision OCR (0.8121 vs. 0.8121, p_{Holm} = 1.0) or degraded blur (0.6667 vs. 0.6684, p_{Holm} = 1.0). Surface features cannot predict representation efficacy under real-world proxy distortions.

B. Exploratory Multi-Hop Query Decomposition

Across diagnostic multi-hop queries (Table XIV), multi-hop LLM bridging beat single-stage hybrid retrieval on 2 queries and lost on 1 query (net +1 hit / 50; Target Hit@10 = 0.640 vs. 0.620, exact McNemar p = 1.0000), matching single-stage dense retrieval (identical 32/50 hits). While LLM bridging outperforms sparse BM25 alone (+8 net hits; p = 0.0215), it incurs a 45% latency penalty over the end-to-end multi-hop pipeline (4,736 ms vs. 3,260 ms, dominated by multi-hop LLM API network roundtrips and autoregressive decoding over two retrieval hops, distinct from the local single-stage lookup latencies in Table XII). Grounded extractive pseudo-relevance feedback achieves identical Hit@10 (0.640 vs. 0.640, exact McNemar p = 1.0000, paired difference 0.0000 [95% CI: $-0.0792, +0.0792$]). Expanding the evaluation to N = 100 relational multi-hop queries resolves this power limitation: on the initial 50 pure 2-hop bridging queries, extractive PRF and generative LLM both achieve Hit@10 = 0.6400 (32/50), whereas on the added 50 relational queries both achieve Hit@10 = 1.0000 (50/50), yielding pooled Hit@10 = 0.8200 (82/100, paired difference 0.0000 [95% CI: $-0.0392, +0.0392$], establishing formal statistical equivalence under TOST at margin $\Delta = \pm 0.05$ with $p_{\text{TOST}} = 0.0062 < 0.01$). While generative bridging incurs a 394× latency multiplier over isolated local retrieval, it yields zero empirical retrieval gain over extractive entity expansion. Dual-reviewer human auditing across 110 cases confirmed high inter-rater agreement and calibration against automated `gpt-4o-mini` scoring: observed agreement of 95.5% for faithfulness (κ = 0.8716), 96.4% for relevance (κ = 0.8926), and 98.2% for correctness (κ = 0.9451; mean κ = 0.9031 [48]).

C. Dual-Tier Feasibility Pilots and Systemic Extensions

Prior to establishing the frozen benchmark, exploratory pilots evaluated multimodal proxies across text (N = 334) and physical media (N = 151 probes: N = 85 scene text, N = 13 ESC-50 audio recordings [49] via LAION-CLAP [41], and N = 53 album photos via CLIP ViT-B/32 [17]). On physical media, sparse BM25 achieved MRR = 0.7916 on scene text, CLAP achieved MRR = 1.0000 on audio, and CLIP achieved Hit@10 = 0.792 (MRR = 0.6182) on raw photos, confirming modality-specific proxy bounds.

Cognitive and Architectural Extensions for Personal Multimodal RAG: Extending this framework to lifelong personal memory vaults reveals four key architectural principles: (1) *Episodic-to-Semantic Consolidation:* In a 2-year longitudinal vault (N = 1,000 episodes), flat unconsolidated RAG collapses on recurrent habits (MRR = 0.2006); our dual-store consolidation architecture (episodic buffer + semantic knowledge graph) achieves 0.9917 MRR overall with 3.95× serving acceler-

TABLE XIII. Empirical Decision Guide for MM-RAG System Builders: Optimal Retrieval Architecture and Statistically Indistinguishable Set ($p > 0.05$) by Regime.

Target Modality	Optical Condition	Query Regime	Optimal Architecture	Metric [95% CI]*	Statistically Tied Set ($p > 0.05$)	Architectural Recommendation
Document Page	Clean Neural OCR	Verbatim Lookup	**Char 3–5g BM25**	0.9436 [0.919, 0.965]	Char BM25 standalone	Sub-word n-grams capture numeric tokens and vendor fragments with zero GPU overhead.
Document Page	Degraded Optical Blur	Verbatim Lookup	**Char 3–5g BM25**	0.8248 [0.782, 0.864]	Char BM25 standalone	Character n-grams tolerate character substitutions and optical blur; word BM25 collapses to 0.6972.
Document Page	Open-Source Tesseract	Verbatim Lookup	**Char 3–5g BM25**	0.8412 [0.803, 0.878]	Char BM25 standalone	Standardized Tesseract (CER 29.4%, line-aligned 24.2%) supports high-precision retrieval without proprietary OCR APIs.
Document Page	Clean Neural OCR	Light Paraphrase	**ColQwen2-v1.0 (GPU)**	0.4962 [0.448, 0.547]	None (ColQwen2 is decisive; $p_{\text{Holm}} = 0.0095$)	Native visual retrieval provides +0.0628 lift over grid Word BM25 ($p_{\text{Holm}} = 0.0072$) and +0.0627 over Tuned Hybrid ($p_{\text{Holm}} = 0.0095$); Tuned Hybrid (0.4335) serves as zero-GPU fallback.
Document Page	Apple Clean Real-OCR	Answerable Question	**Char 3–5g BM25**	0.9328 [0.906, 0.957]	Char BM25 standalone	Real-OCR Char BM25 (0.9328) significantly outperforms Word BM25 (0.8841, diff +0.0487, $p = 0.0018$) and ColQwen2 (0.8595; diff +0.0733, $p = 0.00012$; Clean GT 0.9471 vs. 0.8595, diff +0.0876); 249 ms T4 latency vs. 0.87 ms CPU verbatim (5.72 ms answerable).
Document Visual Form	Visual Layout	Visual Question Answering	**ColQwen2-v1.0 (GPU)**	0.6450 [0.601, 0.689]	ColQwen2 standalone	On DocVQA, native visual retrieval substantially outperforms text proxies (≤ 0.3507) where layout geometry governs answerability.
Natural Image	Unannotated Pixels	Descriptive Search	**SigLIP Base Patch16**	0.6071 [0.584, 0.633]	None (SigLIP is decisive; dominates CLIP and proxies)	Bypass text proxies completely; native visual embeddings dominate baseline BLIP (0.3728) and detailed Florence-2 (0.4420).
Multi-Hop Query	Precomputed Proxies	Relational Multi-Hop	**Extractive PRF Bridge**	0.8200 [0.742, 0.898]†	Extractive PRF (0.820), Generative LLM (0.820)	Extractive entity PRF matches generative LLM bridging on $N = 100$ relational queries ($p_{\text{TOST}} = 0.0062$ at $\Delta = \pm 0.05$), eliminating 394× latency multiplier.

*Metric is MRR for rows 1–7; †Target Hit@10 on $N = 100$ relational queries ($N = 50$ pure 2-hop: 0.6400). Note: Comparisons with no statistically significant difference detected ($p > 0.05$) reflect failure to reject the null hypothesis; they do not assert statistical equivalence.

TABLE XIV. Exact paired discordant case analysis across 50 diagnostic multi-hop queries (MM-RAG Target Hit@10 = 0.640).

Baseline	Base Hit@10	Paired Diff [95% CI]	Disc. Pairs	McN. p	Verdict
Single Hybrid	0.620	+0.020 [-0.04, 0.10]	2 / 1	1.0000	No diff.
Single Dense	0.640	+0.000 [-0.08, 0.08]	2 / 2	1.0000	Tie
Sparse Alone	0.480	+0.160 [+0.04, 0.28]	9 / 1	0.0215	$p < 0.05$

ation (6.2 ms vs. 24.5 ms). (2) *Adaptive Compute Budgeting:* Across $N = 300$ queries stratified by relational complexity, an Adaptive Complexity Router (ACR) achieves 1.0000 MRR under a 15 ms budget, slashing query latency by 41.8% (16.3 ms vs. 28.0 ms full pipeline). (3) *Temporal Logic Planning:* Standard dense search suffers an 82.67% Chronological Violation Rate (CVR) on relative operators ("BEFORE", "AFTER", "DURING"); enforcing interval constraint kernels via a Temporal Logic Planner delivers 0.7666 MRR and 100% constraint satisfaction (0.0% CVR, 3.1 ms latency). (4) *Consent-Aware Privacy Governance:* In multi-user collaborative memory sharing, unconstrained retrieval incurs a 23.0% privacy exposure rate, while naive blanket redaction collapses generation coherence (94.6% → 52.0% ANLS); Consent-Aware Privacy Filtering (CAPF) guarantees 0.00% unconsented exposure while sustaining 93.56% coherence and 0.3258 MRR.

7. LIMITATIONS AND THREATS TO VALIDITY

To facilitate objective assessment, we document key limitations:

1) **Benchmark Domain Concentration:** Primary document evaluations focus on retail receipts (SROIE, $N = 626$), administrative forms (FUNSD, $N = 100$), DocVQA pages ($N = 500$), and natural scenes (MS-COCO, $N = 1{,}000$). Documents with multi-column prose or rich typography may exhibit differing degradation dynamics.
2) **Cross-Engine and Cross-Domain OCR Generalization:** Replication across Apple Vision, Tesseract 5.5, and EasyOCR 1.7.2 confirms character sub-word resilience generalizes across architectures. On FUNSD administrative forms under clean Apple Vision, Character and Word BM25 are statistically indistinguishable on verbatim queries (0.8433 vs. 0.8364, diff +0.0069, $p > 0.05$), with Character BM25 advantage concentrating under severe blur (0.4725 vs. 0.3412). Under paraphrases, Word BM25 leads on receipts (0.4334 vs. 0.2427), whereas on administrative forms, Charac-

ter BM25 and Dense BGE-base (0.2345 EasyOCR, 0.1834 Apple) match or outperform Word BM25 (0.1775 EasyOCR, 0.1573 Apple) due to repeated ambiguous headers. EasyOCR's near-zero MRR (0.02–0.05) under severe blur represents an optical CRAFT text detection failure threshold (67.8% blank transcriptions) rather than an IR ranking deficit.

3) **Conformal Cascade Calibration:** Under marginal loss $L = \mathbf{1}[\text{not escalated} \wedge Y \neq 1]$, split conformal calibration reports an observed mean test miss risk of 0.0491 at a 78.4% escalation rate (N = 2,754, 200 splits). Empirical split coverage (risk $\leq$ 0.05) is 58.5%, matching an equal-escalation fixed-margin baseline (MRR 0.3579 vs. 0.3577, $p > 0.05$). Entity rewrites experience risk 0.0627 (> 0.05), demonstrating that marginal calibration does not guarantee conditional protection per query type.
4) **Answerable Question Formulation:** Our answerable queries evaluate natural factual queries isolating semantic abstraction from lexical absence; conversational user queries may exhibit broader diversity.
5) **Human-Calibrated Generation Audit:** Evaluation of diagnostic generation relies on dual-reviewer human auditing calibrated against automated `gpt-4o-mini` scoring across 110 sampled cases (κ = 0.8716 faithfulness, 0.8926 relevance, 0.9451 correctness, mean κ = 0.9031 [48]).
6) **Multi-Hop Retrieval Scaling and Equivalence:** Expanding multi-hop evaluation to N = 100 relational queries resolves earlier sample size limits (N = 50), establishing formal statistical equivalence between extractive PRF and generative LLM bridging under TOST (p_{TOST} = 0.0062 at Δ = $\pm$0.05, Hit@10 = 0.8200 both; 0.6400 on initial 50, 1.0000 on added 50). Generative bridging incurs a 394× latency multiplier over isolated local retrieval with zero empirical gain over extractive entity expansion.
7) **Dose-Response Non-Linearity:** Retrieval degradation is non-monotonic across query types: entity-substituting paraphrases trigger severe drops (Apple Vision Δ = 0.4037 MRR; Tesseract Δ = 0.3400), whereas answerable questions preserve key topical anchors and sustain high lexical retrieval (MRR = 0.9471).
8) **Synthetic vs. Optical Noise Divergence:** Character n-gram robustness and dictionary reliability weighting tuned on synthetic permutations fail to translate directly to real optical blur; retrieval pipelines must be validated on physical degradation.
9) **LLM-Generated Query Synthesis Scope and Human Validation:** Validation across `Qwen2.5-7B` (N = 306) and human-authored queries (N = 200, Table XI) shows the sparse reversal (Word 0.4334 > Char 0.2427) is specific to entity-substitution paraphrases that eliminate subword overlap; natural human queries preserve sufficient anchors for Char BM25 (0.5549) to approach Word BM25 (0.5766), while dense BGE MRR recovers from 0.1601 to 0.4909.
10) **Dense and Learned Sparse Model Scope:** We benchmark representative bi-encoders (`all-MiniLM-L6-v2`, `bge-base-en-v1.5`); extending benchmarks to multilingual subwords (BGE-M3 [28]), learned sparse representations (SPLADE [29]), and token-level interaction (ColBERTv2 [30]) represents compelling future work.

8. CONCLUSION AND FUTURE WORK

In this work, we have established that intermediate textual proxies in multimodal retrieval-augmented generation exhibit well-defined, regime-dependent efficacy boundaries rather than uniform success or failure. Our empirical investigation across 1,626 multimodal items reveals three principal architectural insights:

First, lexical inverted indexing over character subwords (n-grams of length 3–5) provides a resilient sparse retrieval foundation, outperforming word tokenization across both commercial neural OCR and open-source Tesseract under heavy optical blur (Apple Clean: 0.9436 vs. 0.8371; Degraded: 0.8248 vs. 0.6972; Tesseract: 0.8412 vs. 0.7364). This dominance reverses under entity-substituting light paraphrases, where Word BM25 (0.4334) surpasses Character BM25 (0.2427). Second, the collapse of dense bi-encoders on document text is associated with a query-span formulation mismatch rather than sequence truncation; structuring queries as natural answerable questions improves BGE-Base MRR from 0.3256 to 0.5263 on identical document text. Third, native visual representations strongly outperform machine caption proxies for natural photographs (SigLIP MRR 0.6071 vs. machine captions 0.3728, $p < 10^{-4}$), approaching 5-reference human gold captions within a narrow statistical margin (0.6071 vs. 0.6210, diff -0.0139, paired p = 0.321; formally equivalent at Δ = $\pm$0.05 with p_{TOST} = 0.0050, 90% CI $[-0.0369, +0.0091]$); on visual document pages (DocVQA), native ColQwen2 substantially outperforms generic text proxies (0.6450 vs. $\leq$ 0.3507); however, on standard receipts with natural answerable queries, CPU lexical retrieval (Char BM25 0.9471) outperforms ColQwen2 (0.8595, p = 0.00012) while imposing 0.87–5.72 ms CPU latency vs. 249.21 ms GPU latency.

In exploratory cascade risk calibration, marginal conformal prediction achieved 58.5% empirical split coverage against the 0.05 nominal risk target (82.0% for high-probability RCPS), demonstrating that marginal calibration does not provide conditional risk coverage under entity shifts and performs comparably to a simple fixed-margin baseline (MRR 0.3579 vs. 0.3577). On expanded relational multi-hop queries (N = 100), extractive PRF is formally equivalent to generative LLM bridging under TOST (Hit@10 0.820 vs. 0.820, $p_{\text{TOST}} = 0.0062 < 0.01$ at Δ = $\pm$0.05) while eliminating 394× LLM latency and API cost. By formalizing the trade-offs between computational overhead and representation fidelity, this work provides an engineering roadmap for efficient, trustworthy multimodal retrieval in real-world computing systems.

References

[1] Z. Jing et al., "When large language models meet vector databases: A survey," *arXiv preprint arXiv:2402.01763*, 2024.
[2] P. Lewis et al., "Retrieval-augmented generation for knowledge-

intensive NLP tasks," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 33, pp. 9459–9474, 2020.

[3] M. Abootorabi et al., "Ask in any modality: A comprehensive survey on multimodal retrieval-augmented generation," in *Findings of the Association for Computational Linguistics: ACL 2025*, pp. 16776–16809, 2025.

[4] A. Patel, R. Shivani, N. V. Usha, and S. A, "Enhancing interactive querying with a multimodal RAG system: Integrating text, video, and document analysis via LLaMA3," in *Proc. IEEE ETCC*, 2025.

[5] P. Joshi et al., "Robust multi model RAG pipeline for documents containing text, table & images," in *Proc. IEEE ICAAIC*, pp. 1–7, 2024.

[6] M. Faysse et al., "ColPali: Efficient document retrieval with vision language models," in *Proc. ICLR*, 2025.

[7] P. B. Kantor and E. M. Voorhees, "The TREC-5 Confusion Track: Comparing retrieval methods for scanned text," *Information Retrieval*, vol. 2, no. 2–3, pp. 165–176, 2000.

[8] P. McNamee and J. Mayfield, "Character N-gram tokenization for European language text retrieval," *Information Retrieval*, vol. 7, no. 1–2, pp. 73–97, 2004.

[9] Apple Inc., "VNRecognizeTextRequest: An image-analysis request that finds and recognizes text in an image," *Apple Developer Documentation*, 2020.

[10] R. Smith, "An overview of the Tesseract OCR engine," in *Proc. 9th Int. Conf. on Document Analysis and Recognition (ICDAR)*, IEEE, pp. 629–633, 2007.

[11] JaidedAI, "EasyOCR: Ready-to-use optical character recognition with 80+ supported languages," 2020. [Online]. Available: https://github.com/JaidedAI/EasyOCR.

[12] M. Faysse, H. Massa, A. Poli, and F. Guegan, "ColQwen2-v1.0: Vision-language document retrieval model," Hugging Face Model Hub, 2024. [Online]. Available: https://huggingface.co/vidore/colqwen2-v1.0.

[13] M. Mathew, D. Karatzas, and C. V. Jawahar, "DocVQA: A dataset for VQA on document images," in *Proc. IEEE/CVF WACV*, pp. 2200–2209, 2021.

[14] A. Karpathy and L. Fei-Fei, "Deep visual-semantic alignments for generating image descriptions," in *Proc. IEEE CVPR*, pp. 3128–3137, 2015.

[15] T.-Y. Lin et al., "Microsoft COCO: Common objects in context," in *Proc. ECCV*, pp. 740–755, 2014.

[16] X. Zhai et al., "Sigmoid loss for language image pre-training," in *Proc. IEEE/CVF ICCV*, pp. 11975–11986, 2023.

[17] A. Radford et al., "Learning transferable visual models from natural language supervision," in *Proc. ICML*, pp. 8748–8763, 2021.

[18] R. Girdhar et al., "ImageBind: One embedding space to bind them all," in *Proc. IEEE/CVF CVPR*, pp. 15180–15190, 2023.

[19] S. Robertson and H. Zaragoza, "The probabilistic relevance framework: BM25 and beyond," *Foundations and Trends in Information Retrieval*, vol. 3, no. 4, pp. 333–389, 2009.

[20] S. E. Robertson, S. Walker, S. Jones, M. M. Hancock-Beaulieu, and M. Gatford, "Okapi at TREC-3," *NIST Special Publication 500-225*, pp. 109–126, 1995.

[21] J. Cho, D. Mahata, O. İrsoy, Y. He, and M. Bansal, "M3DocRAG: Multi-modal retrieval is what you need for multi-page multi-document understanding," *arXiv preprint arXiv:2411.04952*, 2024.

[22] P. Wang et al., "Qwen2-VL: Enhancing vision-language models with dynamic resolution," *arXiv preprint arXiv:2409.12191*, 2024.

[23] A. Fateh, V. Abolghasemi, and G. Anbarjafari, "A comprehensive review of optical character recognition (OCR) systems, *Engineering Reports*, vol. 6, no. 6, p. e12833, 2024.

[24] N. Otsu, "A threshold selection method from gray-level histograms," *IEEE Transactions on Systems, Man, and Cybernetics*, vol. 9, no. 1, pp. 62–66, 1979.

[25] P. V. Hough, "Method and means for recognizing complex patterns," U.S. Patent 3,069,654, 1962.

[26] N. Reimers and I. Gurevych, "Sentence-BERT: Sentence embeddings using Siamese BERT-networks," in *Proc. EMNLP*, pp. 3982–3992, 2019.

[27] S. Xiao, Z. Liu, P. Zhang, and N. Muennighoff, "C-Pack: Packaged resources to advance general Chinese embedding," *arXiv preprint arXiv:2309.07597*, 2023.

[28] J. Chen et al., "BGE M3-Embedding: Multi-lingual, multi-functionality, multi-granularity text embeddings through versatile data curation," *arXiv preprint arXiv:2402.03216*, 2024.

[29] T. Formal, B. Piwowarski, and S. Clinchant, "SPLADE: Sparse lexical and expansion model for first stage ranking," in *Proc. 44th ACM SIGIR*, pp. 2288–2292, 2021.

[30] K. Santhanam, O. Khattab, J. Saad-Falcon, C. Potts, and M. Zaharia, "ColBERTv2: Effective and efficient retrieval via lightweight late interaction," in *Proc. NAACL*, pp. 3715–3734, 2022.

[31] O. Khattab et al., "Baleen: Robust multi-hop reasoning at scale via condensed retrieval," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 34, pp. 27670–27682, 2021.

[32] L. Wang, J. Lin, and D. Metzler, "A cascade ranking model for efficient ranked retrieval," in *Proc. 34th Int. ACM SIGIR Conf. on Research and Development in Information Retrieval (SIGIR)*, pp. 105–114, 2011.

[33] L. Gallagher, R.-C. Chen, R. Blanco, and J. S. Culpepper, "Joint optimization of cascade ranking models," in *Proc. 12th ACM Int. Conf. on Web Search and Data Mining (WSDM)*, pp. 15–23, 2019.

[34] S. Bates, A. Angelopoulos, L. Lei, J. Malik, and M. I. Jordan, "Distribution-free risk-controlling prediction sets," *Journal of the ACM*, vol. 68, no. 6, pp. 1–34, 2021.

[35] A. N. Angelopoulos, S. Bates, M. I. Jordan, and L. Lei, "Conformal risk control," in *Proc. 12th Int. Conf. on Learning Representations (ICLR)*, 2024.

[36] A. N. Angelopoulos and S. Bates, "A gentle introduction to conformal prediction and distribution-free uncertainty quantification," *arXiv preprint arXiv:2107.07511*, 2021.

[37] J. Li et al., "BLIP: Bootstrapping language-image pre-training for unified vision-language understanding and generation," in *Proc. ICML*, pp. 12888–12900, 2022.

[38] J. Li, D. Li, S. Savarese, and S. Hoi, "BLIP-2: Bootstrapping language-image pre-training with frozen image encoders and large language models," in *Proc. ICML*, pp. 19730–19742, 2023.

[39] B. Xiao et al., "Florence-2: Advancing a unified representation for versatile vision tasks," in *Proc. IEEE/CVF CVPR*, 2024.

[40] A. Radford et al., "Robust speech recognition via large-scale weak supervision," in *Proc. ICML*, pp. 28492–28518, 2023.

[41] Y. Wu et al., "Large-scale contrastive language-audio pretraining with feature fusion and keyword tagging," in *Proc. IEEE ICASSP*, pp. 1–5, 2023.

[42] G. V. Cormack, C. L. Clarke, and S. Büttcher, "Reciprocal rank fusion outperforms Condorcet and individual rank learning methods, in *Proc. 32nd Int. ACM SIGIR Conf. on Research and Development in Information Retrieval*, pp. 758–759, 2009.

[43] R. Nogueira and K. Cho, "Passage re-ranking with BERT," *arXiv preprint arXiv:1901.04085*, 2019.

[44] Z. Huang et al., "ICDAR 2019 competition on scanned receipt OCR and information extraction (SROIE)," in *Proc. 15th ICDAR*, pp. 1516–1520, 2019.

[45] G. Jaume, H. K. Ekenel, and J.-P. Thiran, "FUNSD: A dataset for form understanding in noisy scanned documents," in *Proc. ICDAR Workshops*, pp. 1–6, 2019.

[46] OpenAI, "GPT-4o mini: Advancing cost-efficient intelligence, *OpenAI Blog*, July 18, 2024. [Online]. Available: https://openai.com/index/gpt-4o-mini-advancing-cost-efficient-intelligence/.

[47] N. Vaessen, "JiWER: Similarity measures for automatic speech recognition evaluation," *Python Package Index*, 2024.

[48] L. Zheng et al., "Judging LLM-as-a-judge with MT-Bench and Chatbot Arena," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 36, pp. 46595–46623, 2023.

[49] K. J. Piczak, "ESC: Dataset for environmental sound classification," in *Proc. 23rd ACM Int. Conf. on Multimedia (ACM MM)*, pp. 1015–1018, 2015.